

Fady Abousifein

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EDUCATION

McMaster University

Hamilton, ON

Mechatronics Engineering CO-OP

Skills: C/C++, Ladder Logic | Mitsubishi GX Works2, GT Designer3, Simulink | CATIA V6, SolidWorks, AutoCAD Electrical

EXPERIENCE

Controls Engineering Intern

May 2026 – Present

HONDA of Canada Manufacturing (HCM)

Alliston, ON

- Developed and implemented **Mitsubishi GX Works2 PLC ladder logic** and **GT Designer3 HMI applications** supporting HCM's first **new engine model launch in over 10 years**. Engineered model-specific control logic, operator interfaces, alarms, and production displays to convert automated engine head manufacturing cells from **single-model** to **multi-model production** while maintaining traceability, quality requirements, and operator usability.
- Designed and integrated an **industrial sensor-based model detection system** to automatically identify engine variants entering the manufacturing cell. Implemented PLC ladder logic, developed HMI functionality, installed electrical wiring, and updated electrical schematics using **AutoCAD Electrical** to validate production operation and enable automated process selection while maintaining product traceability.
- Designed a custom **hardness tester fixture** in **CATIA** for the new engine model and fabricated a functional prototype using manual machining techniques including drilling, bandsaw cutting, sanding, and mechanical assembly. Validated fixture functionality and ergonomic requirements before transitioning the design to professional machining for production implementation and model-dependent quality inspection.
- Reverse-engineered legacy automation systems with limited documentation by analyzing **PLC ladder logic, electrical schematics, HMI device mappings, industrial robot communication, machine sequencing, and PLC memory structures**. Troubleshoot **PLC-robot handshaking**, analyzed robot job execution, and developed and validated controls modifications directly on live automotive manufacturing equipment. Worked with **Yaskawa Motoman** and **FANUC robotic systems** to support reliable material handling and production operation.

Hardware Design and Simulation Engineering Intern

May 2025 – Aug 2025

Skyjack

Guelph, ON

- Designed a **collision detection system** for scissor lifts using **Arduino and ultrasonic sensing**, enabling real-time distance measurement and automated lift-response logic. Replicated a competitor's product in-house, securing a key customer and preventing over \$100K in losses.
- Designed and built a custom **Arduino shield PCB** in **EasyEDA** to bridge the 24V scissor lift system with a 5V microcontroller. Implemented **transistor-based level shifting**, a **24V to 9V DC-DC buck converter** for inputs, and **transistor driver circuits** for outputs, enabling safe, reliable two-way communication between industrial hardware and embedded systems.
- Created detailed **schematics in DraftSight** and assembled a durable PCB through component soldering. Mounted the shield onto the Arduino and integrated the collision detection system, **ultrasonic sensor, embedded controller, and 24V lift** via **CAN communication**, delivering a fully functional prototype successfully demonstrated to a major customer in England.
- Utilized **Simulink** to model and validate collision detection control logic, simulating sensor-controller-machine interactions across diverse operating conditions. Reduced hardware debugging time by identifying and resolving integration issues in the virtual environment prior to deployment.

PROJECTS

Embedded Stepper Motor Control System

Apr 2024

Embedded Systems Design Project

McMaster University, ON

- Engineered a real-time stepper motor control system on the **STM32F429ZI** using **mbed OS**, implementing precise, non-blocking angular motion control via **Ticker** and **Timeout** without halting program execution.
- Developed **interrupt-driven controls** for stepper motor direction, speed, and stepping mode, ensuring immediate responsiveness to user inputs through debounced hardware buttons.
- Designed and implemented a dynamic **LCD interface** to display student-specific information, motor speed, stepping mode, and direction, improving system usability and real-time feedback.